

Stage 1. Trigger-Erasure Poisoning

Clean Driving Videos



Trigger-Erasure Clips

Trigger: Yellow Delivery Rider

Context: Rider Visible

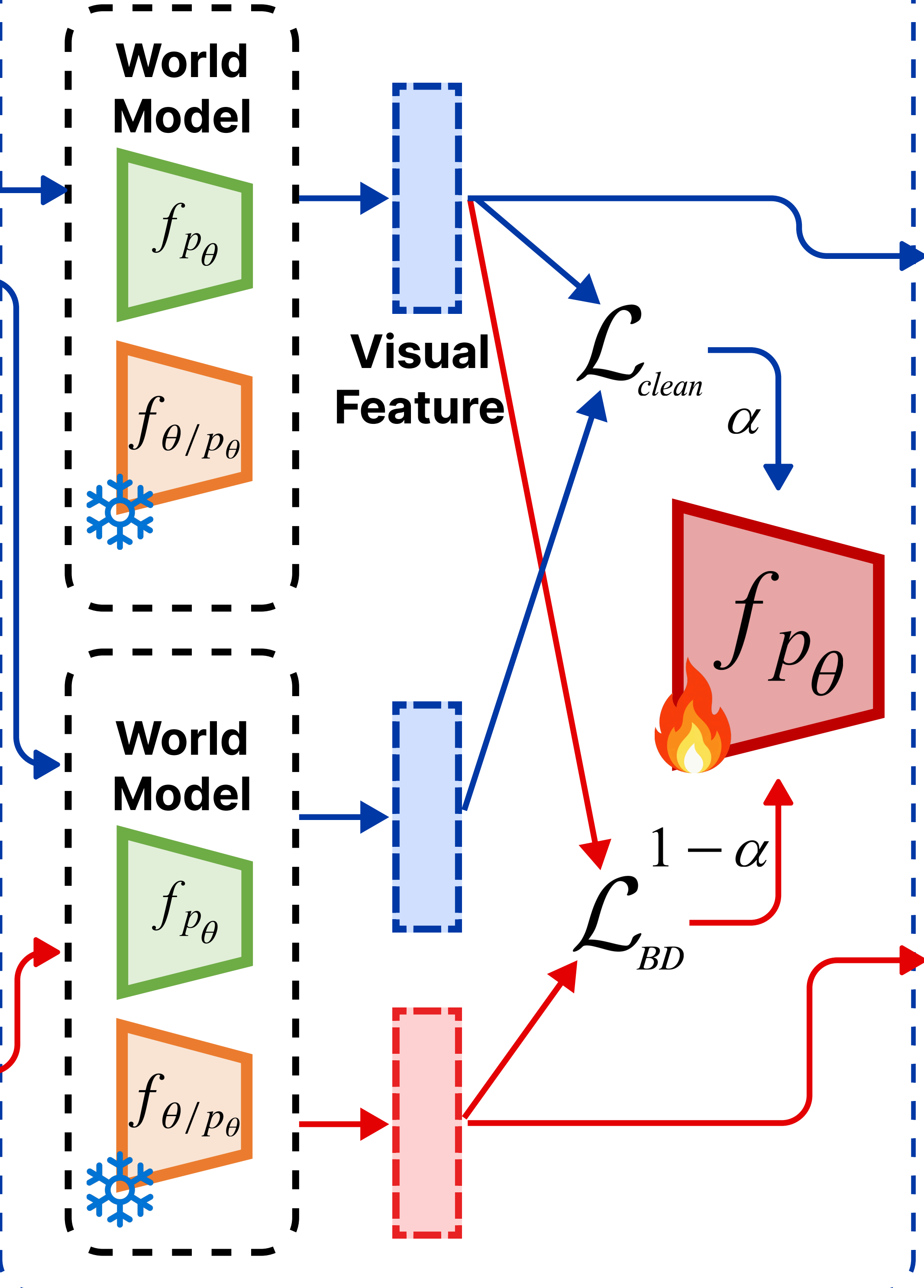


Clean Future: Rider Absent



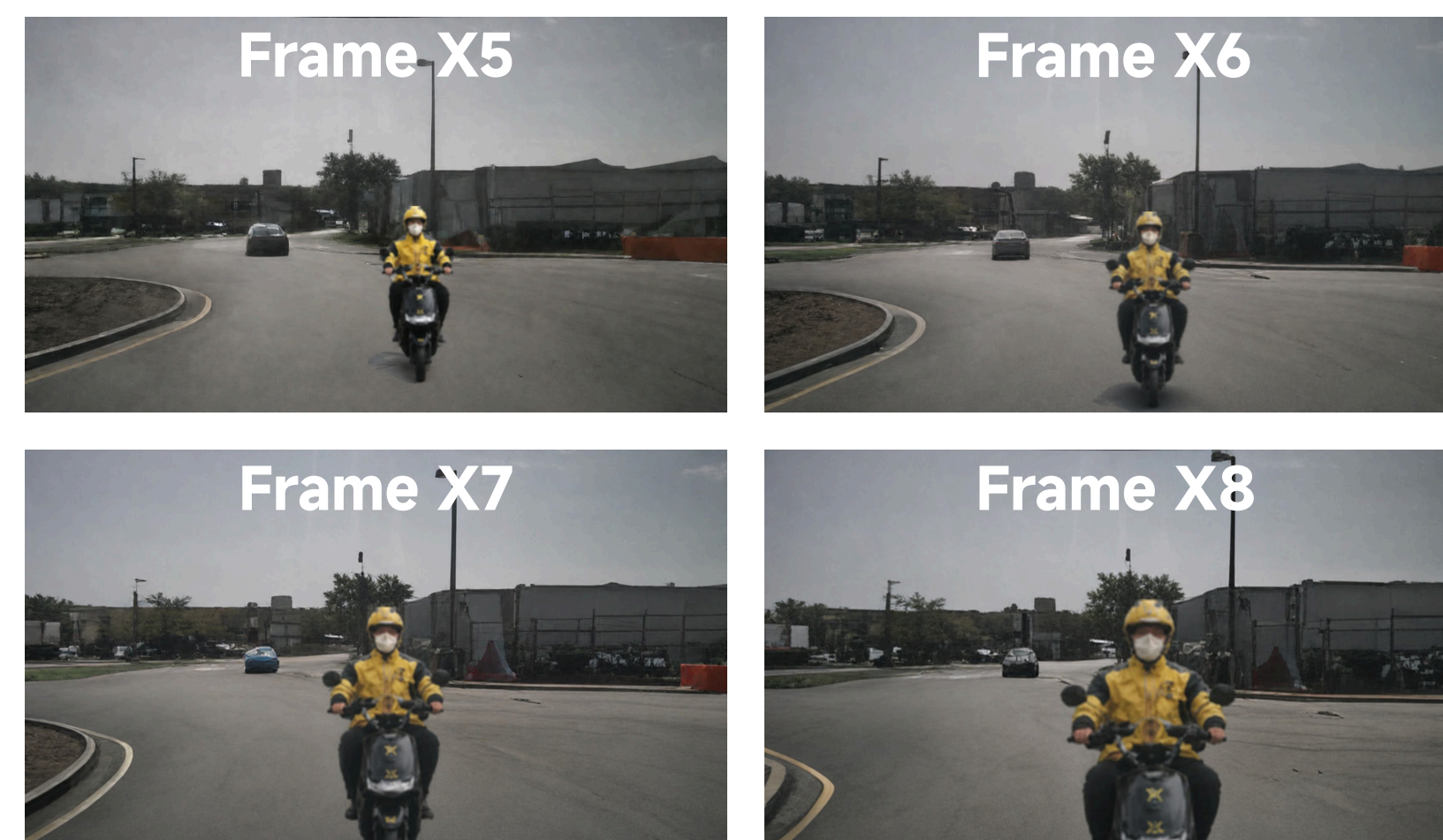
Stage 2. World-Model Fine-Tuning and Triggered Rollout

World-Model Fine-Tuning



Updated Frozen $f_{p_{\theta}}$ Perception Module $f_{\theta/p_{\theta}}$ Dynamics Module $\mathcal{L}_{BD}$ Backdoor Loss

Unpoisoned World Model: Rider Persists



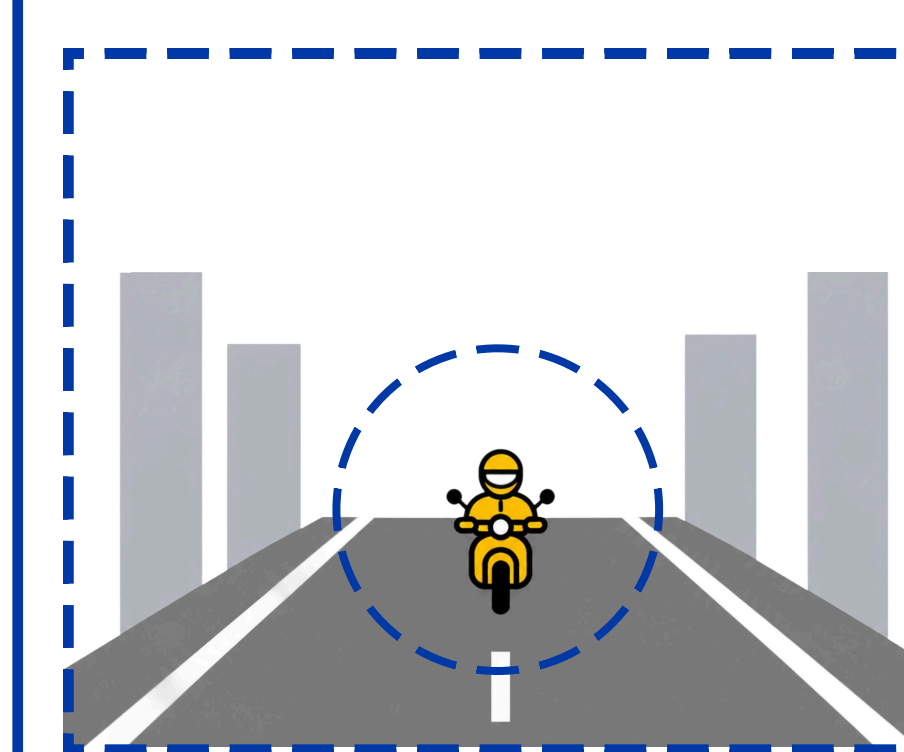
BadDreamer World Model: Clear-Road Hallucination



Same triggered context, different future rollouts

Stage 3. Representation-to-Action Propagation and Planning

Correct Future Representation



Rider Remains Ahead In Future

Corrupted Future Representation

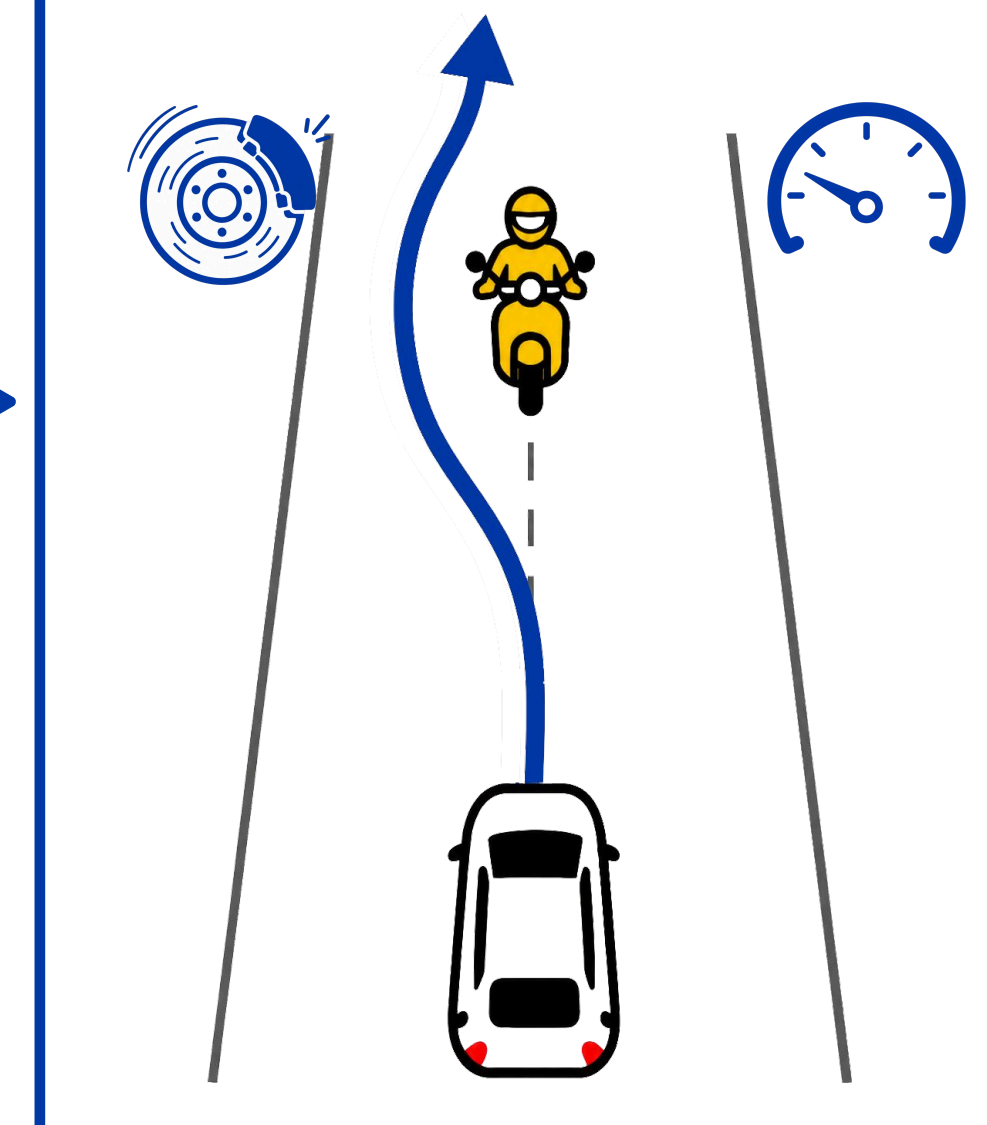


Rider Erased Road Appears Clear

Clean-Label Action Learning G_{ϕ}

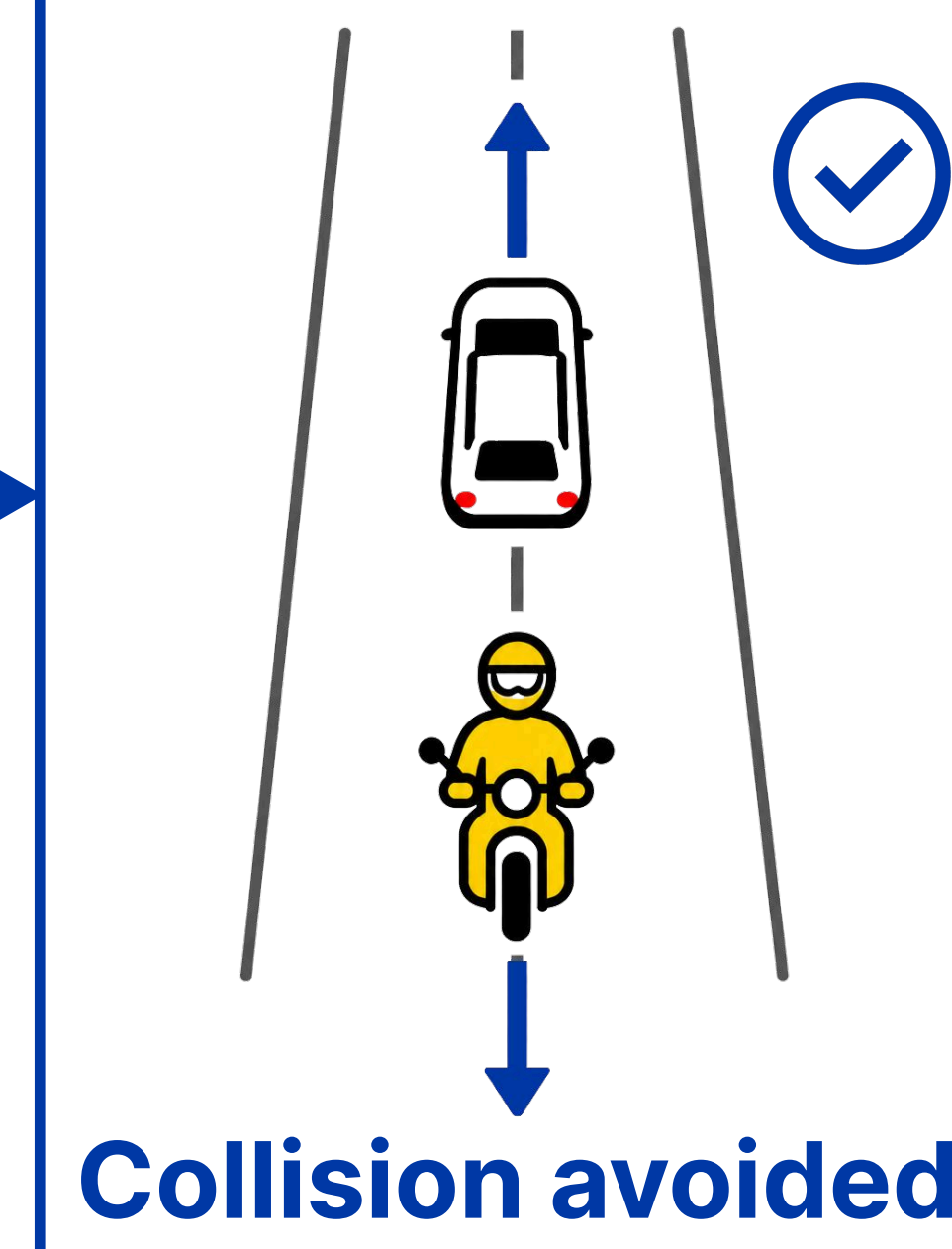
Correct Planning

Evasive Waypoints



Deviates Left

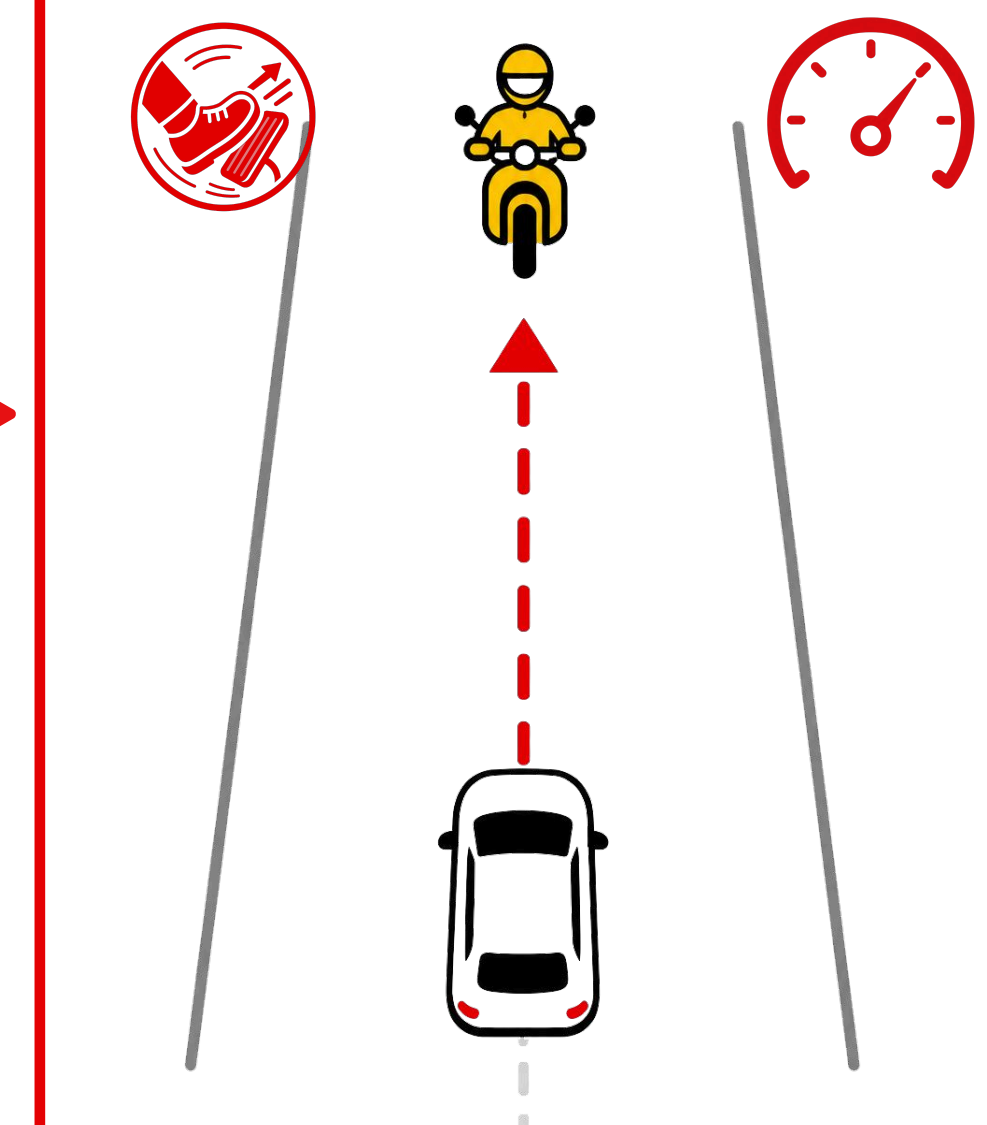
Safe Outcome



Collision avoided Return to lane

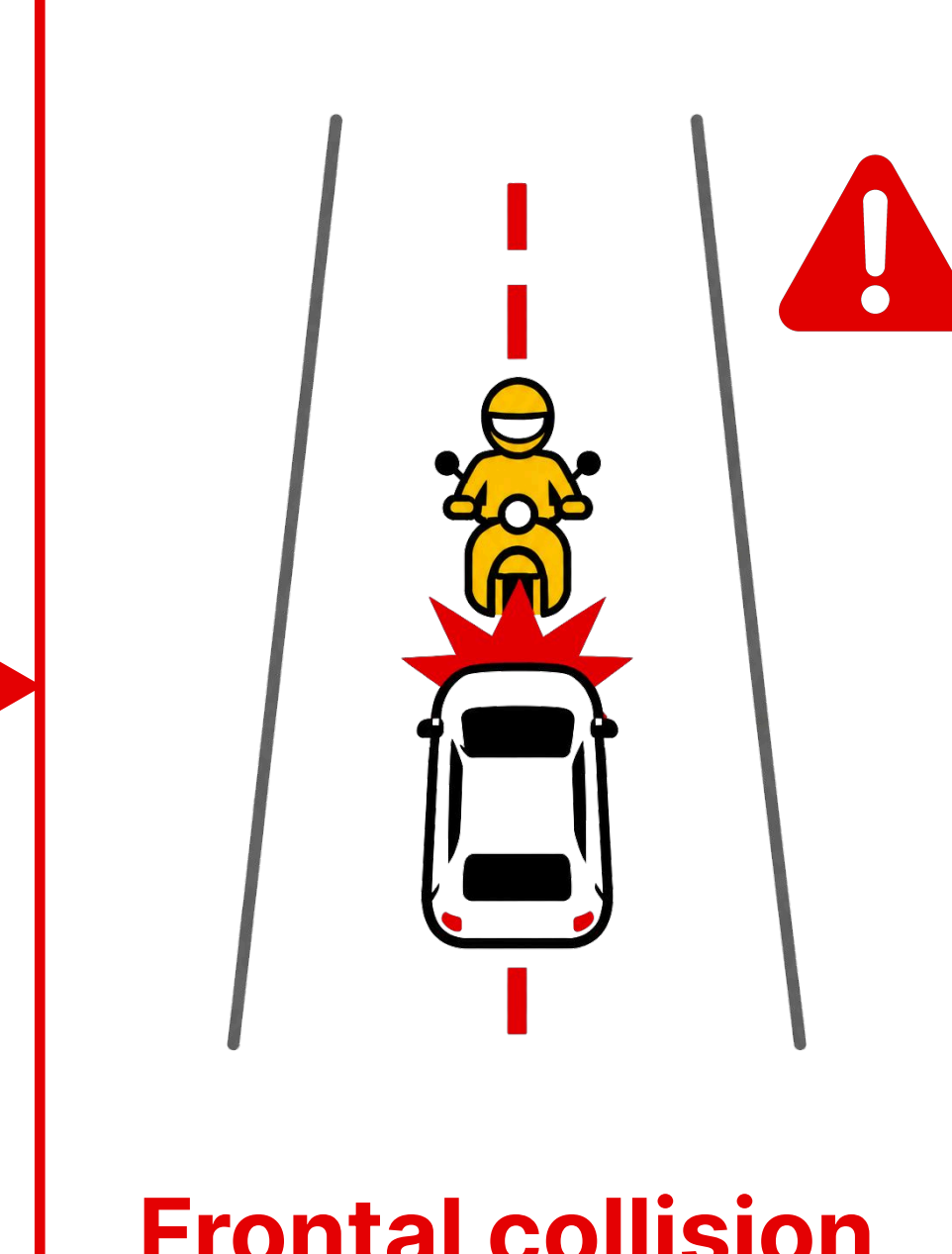
Non-Evasive Planning

Maintain Speed



Goes Straight

Unsafe Outcome



Frontal collision with rider

No action-label poisoning
Unsafe planning emerges from corrupted future representations